WEEK 1 ASSIGNMENT | 100 MARKS

Cyber Security Fundamentals & Network Basics

Project Report

Case study organisation: **An E-commerce Platform (Amazon.in used as the reference example)**

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Note: This report uses a publicly known e-commerce platform only as an illustrative case study. All security discussion is educational and based on general industry concepts, not on any confidential information of the organisation.

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Marks Mapping

Task	Topic	Marks
1	Introduction to Cyber Security	20
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	Total	**100**

Chosen Organisation and Scope

For this assignment, an **online e-commerce platform** (using Amazon.in as the reference example) has been chosen. An e-commerce site is an ideal case study because it combines almost everything a security student needs to learn: web servers, customer databases, online payments, user accounts, public Internet exposure, and constant attention from attackers.

Type	Online marketplace / retail website and mobile app
Assets to protect	Customer personal data (name, address, phone, email), login credentials, payment information, order history, seller data, website availability, brand reputation
Users	Customers, sellers, delivery partners, employees, administrators
Why security matters	A single breach or outage can cause financial loss, legal penalties, and loss of customer trust.

Throughout the report, each concept is first explained in general terms and then linked back to this e-commerce platform.

Task 1: Introduction to Cyber Security

1.1 Definition of Cyber Security

Cyber security is the practice of protecting computers, networks, software, and data from theft, damage, unauthorised access, and disruption. In simple words, it is like putting locks, alarms, and guards around our digital world so that only the right people can use the right information at the right time. It covers technology (such as firewalls and encryption), processes (such as incident response), and people (such as awareness training).

Example: When you shop online, cyber security ensures that your password is not stolen, your card details are not read by a hacker, and the website stays available when you want to buy something.

1.2 Importance of Cyber Security

- **Protects sensitive data:** Personal, financial, and business data is valuable and is a prime target for criminals.
- **Prevents financial loss:** Fraud, ransom demands, and downtime cost organisations heavily.
- **Maintains trust:** Customers only use platforms they believe are safe. A breach damages brand reputation.
- **Ensures business continuity:** Attacks such as DoS can stop an online store from operating, causing lost sales.
- **Meets legal requirements:** Laws and standards (for example India's IT Act 2000, the DPDP Act 2023, and PCI-DSS for card payments) require organisations to protect data.
- **Protects national security:** Power grids, banks, transport, and government services all depend on secure digital systems.
- **Protects individuals' privacy:** Prevents identity theft, stalking, and misuse of personal information.

1.3 Goals of Cyber Security: The CIA Triad

The CIA Triad is the core model of information security. Every security control is designed to support one or more of its three goals.

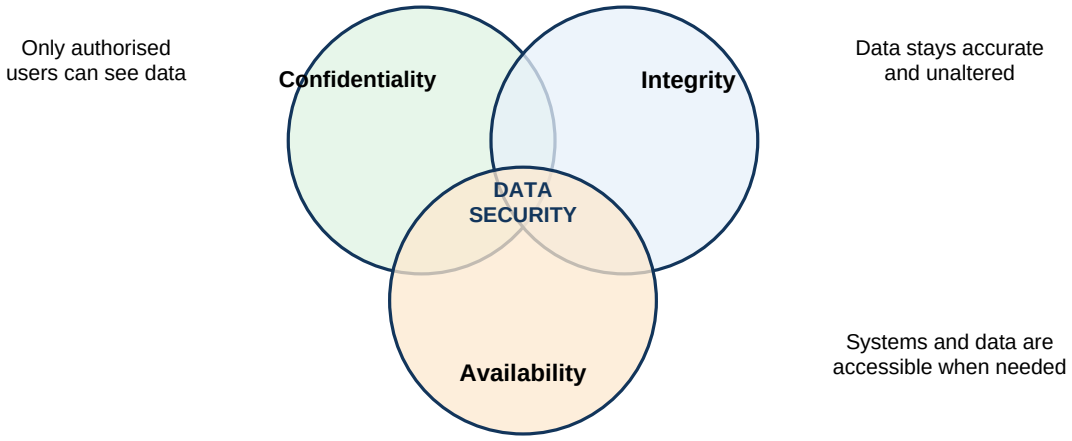

Figure 1: The CIA Triad

Goal	Meaning	E-commerce Example	Controls
Confidentiality	Only authorised people can access information.	Only the customer can see their saved addresses and cards.	Passwords, encryption, access control, MFA
Integrity	Data remains accurate, complete, and unchanged by unauthorised people.	An attacker cannot change the price of a product or the delivery address of an order.	Hashing, digital signatures, checksums, audit logs
Availability	Systems and data are accessible whenever needed.	The site stays online during a big sale event.	Backups, redundancy, load balancers, DoS protection

1.4 Five Real-Life Applications of Cyber Security

#	Area	How cyber security is applied
1	**Online banking & payments (UPI, cards)**	OTP/MFA, end-to-end encryption, fraud detection, and tokenisation protect money transfers.
2	**E-commerce**	HTTPS, secure payment gateways, bot detection, and account protection keep shoppers and sellers safe.
3	**Healthcare**	Patient records are encrypted and access-controlled to protect privacy and keep hospital systems running.
4	**Government & defence**	Protects national databases, elections, citizen services (such as Aadhaar-linked systems), and critical infrastructure.
5	**Education & cloud/remote work**	Secures online classes, student records, exams, and company data accessed via VPN and cloud services.

Task 2: Types of Cyber Threats

A cyber threat is any possible attack that can harm or steal data from a system. The seven threats listed in the assignment (the minimum required is five) are explained below, each with its definition, working, and prevention, along with a short example related to an e-commerce platform.

Threat	One-line summary
Malware	Umbrella term for any malicious software
Virus	Malware that attaches to files and spreads when they are run
Ransomware	Locks or encrypts data and demands payment
Phishing	Fake messages that trick users into revealing information
Spyware	Secretly monitors and steals user activity
Trojan	Harmful program disguised as a useful one
DoS Attack	Floods a service to make it unavailable

2.1 Malware

Definition	Malware (malicious software) is any program or code intentionally created to harm, exploit, or gain unauthorised access to a computer, network, or data. Viruses, ransomware, spyware, and trojans are all types of malware.
How it works	It enters through infected email attachments, pirated software, malicious websites, USB drives, or fake app downloads. Once installed it can steal data, delete files, give attackers remote control, or slow down the system.
Prevention	Install and update antivirus/anti-malware; keep OS and apps patched; download software only from official sources; do not open unknown attachments; use least-privilege accounts; take regular backups.
E-commerce example	Malware on an employee's laptop could give attackers a path to internal order-management systems.

2.2 Virus

Definition	A computer virus is a type of malware that attaches itself to a legitimate file or program and replicates when that file is executed, spreading to other files and systems.
How it works	The user runs an infected file. The virus copies its code into other programs or documents and may corrupt data, crash the system, or spread via email and shared drives. It needs a host file and human action to spread. (A famous example is the ILOVEYOU virus of 2000.)
Prevention	Use updated antivirus with real-time scanning; avoid pirated software and unknown files; scan USB drives; disable macros in documents from unknown senders; keep backups.
E-commerce example	A seller uploads an infected product spreadsheet; if opened by staff, it could spread across office computers.

2.3 Ransomware

Definition	Ransomware is malware that encrypts a victim's files (or locks the entire system) and demands a ransom, usually in cryptocurrency, in exchange for the decryption key.
How it works	It arrives through phishing emails, malicious downloads, or unpatched vulnerabilities. It silently encrypts files, then displays a ransom note with a deadline. Some variants also steal data and threaten to leak it (double extortion). The 2017 WannaCry attack affected hundreds of thousands of computers worldwide.
Prevention	Maintain offline/off-site backups and test restores; patch systems promptly; use email filtering; restrict user privileges; segment the network; train staff; do not pay the ransom without expert and legal advice.
E-commerce example	If ransomware locks the order database, the platform cannot process orders until systems are restored.

2.4 Phishing

Definition	Phishing is a social-engineering attack in which criminals send fake emails, SMS (smishing), calls (vishing), or websites that look genuine, to trick people into revealing passwords, OTPs, or card details.
How it works	The attacker impersonates a trusted brand: for example "Your order is on hold, click here to update payment". The link opens a fake login page that captures the victim's credentials, which the attacker then uses to take over the account.
Prevention	Check the sender's address and URL carefully; never share OTPs or passwords; do not click suspicious links; enable multi-factor authentication; use email spam filters; report phishing messages; go to the official site by typing the address.
E-commerce example	Fake "refund" or "delivery failed" messages pretending to be from the shopping platform to steal customer logins.

2.5 Spyware

Definition	Spyware is malware that secretly installs on a device to monitor and collect information such as keystrokes, browsing habits, passwords, and screen activity, and sends it to an attacker.
How it works	It is bundled with free software or installed through malicious links. It runs silently in the background, records activity (keyloggers, screen capture), and uploads the data to a remote server. Symptoms include slow performance, unknown programs, and unusual network activity.
Prevention	Install trusted security software; avoid free software from unknown sources; keep systems updated; review app permissions; use a firewall; run regular scans; use on-screen or password-manager input for sensitive logins.
E-commerce example	A keylogger on a customer's PC could capture their e-commerce password and card number during checkout.

2.6 Trojan

Definition	A Trojan horse is malware that disguises itself as legitimate or useful software but performs harmful actions once the user installs or runs it. Unlike viruses, Trojans do not self-replicate.
How it works	The user is tricked into installing a "free game", "cracked software", or fake app. In the background, the Trojan opens a backdoor, steals data, downloads other malware, or gives the attacker remote control. Banking Trojans such as Zeus have targeted online banking users.
Prevention	Download only from official stores or vendor websites; verify app permissions and reviews; do not install cracked software; use endpoint protection; apply the principle of least privilege; keep systems patched.
E-commerce example	A fake "shopping coupon" app that secretly steals login details and SMS OTPs from the user's phone.

2.7 DoS Attack (Denial of Service)

Definition	A Denial-of-Service attack tries to make a website, server, or network unavailable to its legitimate users by overwhelming it with traffic or requests. When launched from many machines at once it is called a DDoS (Distributed DoS).
How it works	The attacker sends a huge number of requests, often using a botnet of infected devices. The server's bandwidth, CPU, or memory is exhausted, so genuine customers cannot connect or the site becomes extremely slow.
Prevention	Use DoS/DDoS protection services and CDNs; configure rate limiting; use load balancers and scalable cloud infrastructure; deploy firewalls/IPS; monitor traffic for anomalies; keep an incident-response plan.
E-commerce example	A DDoS attack during a festival sale could take the shop offline, causing lost revenue and customer frustration.

Task 3: Network Basics

3.1 What is a Computer Network?

A computer network is a group of two or more computing devices (computers, phones, servers, printers) connected together using cables or wireless signals so that they can **share data, resources, and services**. The Internet is the largest network in the world, connecting billions of devices. Networks make communication, file sharing, printer sharing, online shopping, and cloud services possible.

E-commerce link: A shopping platform runs on a network that connects customers' devices, web servers, application servers, databases, and payment gateways.

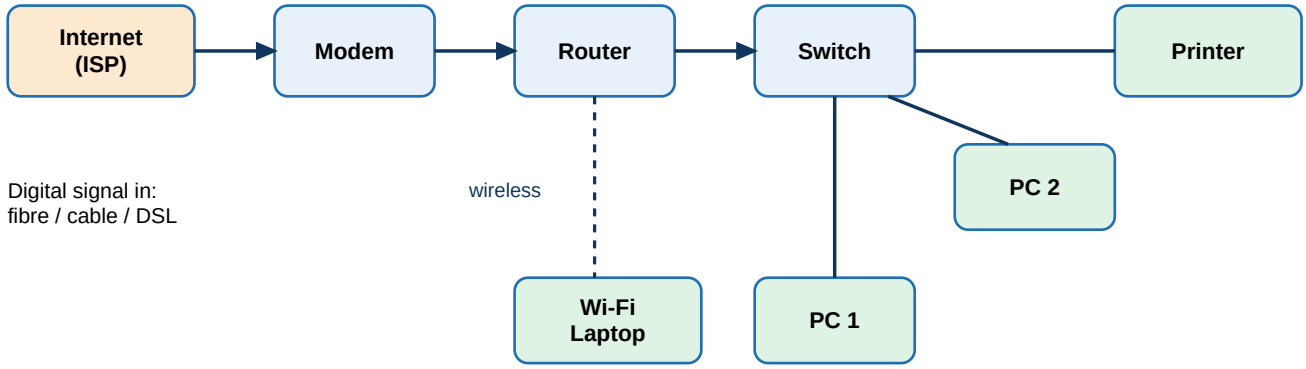

Figure 2: A simple network and how devices connect to the Internet

3.2 Types of Networks

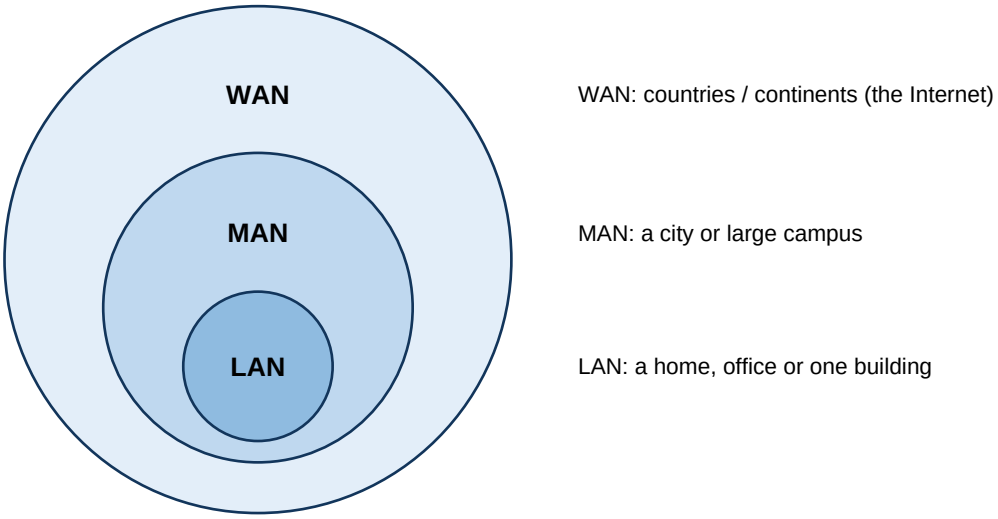

Figure 3: Relative size of LAN, MAN and WAN

Type	Full form	Coverage	Speed / Ownership	Example
LAN	Local Area Network	Small area: room, home, office, or building (up to a few km)	High speed; privately owned	Office Wi-Fi, college computer lab
MAN	Metropolitan Area Network	A city or large campus (about 5 to 50 km)	Medium to high speed; owned by a city or telecom provider	Cable TV network or a city-wide college network

Type	Full form	Coverage	Speed / Ownership	Example
WAN	Wide Area Network	Large geographic area: countries or the whole world	Lower speed, higher cost; uses leased/telecom lines	The Internet; a company linking branches in different cities

3.3 IP Address

An **IP (Internet Protocol) address** is a unique logical address assigned to every device on a network. It identifies the device and its location in the network so that data can be delivered correctly, just like a postal address for a house.

- **IPv4:** 32-bit address written as four decimal numbers (e.g., `192.168.1.10`); about 4.3 billion addresses.
- **IPv6:** 128-bit address written in hexadecimal (e.g., `2001:db8::1`), created because IPv4 addresses are running out.
- An IP address can change when you change networks, so it is a *logical* address.

3.4 MAC Address

A **MAC (Media Access Control) address** is a unique 48-bit hardware address burned into a device's network interface card (NIC) by the manufacturer, written as six pairs of hexadecimal digits (e.g., `00:1A:2B:3C:4D:5E`). It identifies the physical device within a local network and is used at the Data Link layer.

Feature	IP Address	MAC Address
Type	Logical (software-assigned)	Physical (hardware-assigned)
Length	32-bit (IPv4) / 128-bit (IPv6)	48-bit
Changes?	Yes, depends on the network	Normally fixed (can be spoofed)
Used for	Routing data between networks	Delivering data inside a local network
OSI layer	Network layer (Layer 3)	Data Link layer (Layer 2)

3.5 Network Devices: Router, Switch, Modem

Device	What it does	Key points
Router	Connects different networks (e.g., your home LAN to the Internet) and forwards data packets along the best path using IP addresses.	Works at Layer 3; provides NAT, DHCP, Wi-Fi, and basic firewall features; gives the LAN its gateway address.
Switch	Connects multiple devices *within the same LAN* and forwards data only to the intended device using MAC addresses.	Works at Layer 2; more efficient and secure than a hub; creates a star topology.
Modem	Modulates and demodulates signals: converts the ISP's signal (cable/fibre/phone line) into digital data your devices understand, and vice versa.	Connects the home to the ISP; often combined with a router in one box (a gateway).

3.6 DNS (Domain Name System)

The **Domain Name System (DNS)** is the "phonebook of the Internet". Humans remember names like *www.amazon.in*, but computers communicate with IP addresses. DNS translates the domain name into the IP address of the web server, so users do not need to memorise numbers.

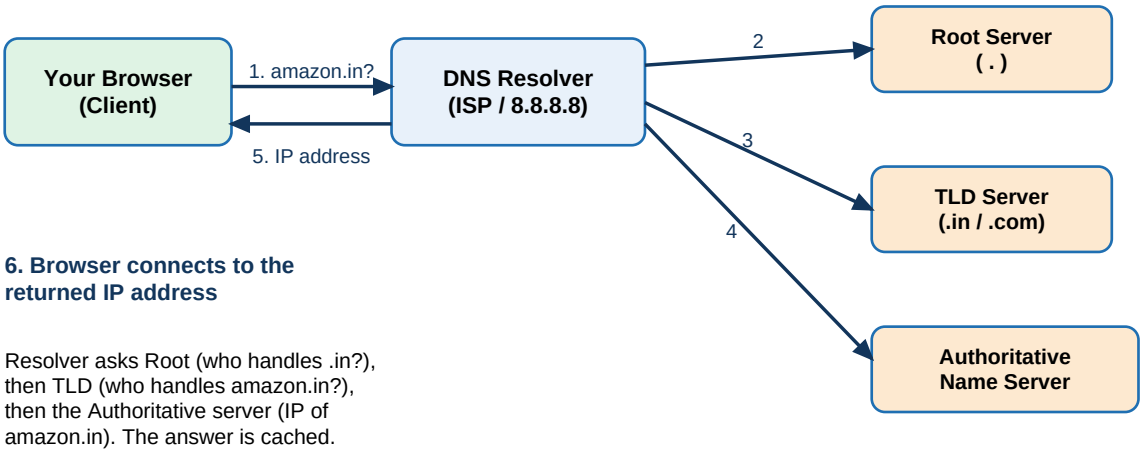

Figure 4: How a DNS lookup resolves a domain name to an IP address

DNS answers are cached by the browser, operating system, and resolver to make later visits faster. DNS can be attacked (DNS spoofing/cache poisoning), so secure DNS options such as DNSSEC and DNS-over-HTTPS are used for protection.

Task 4: Internet Communication Process

This section explains how data travels from a user's computer to a website such as an e-commerce store and back again.

4.1 Key Concepts

Term	Explanation
Client	The device or program that *requests* a service, such as your laptop with a web browser (Chrome, Firefox).
Server	A powerful computer that *provides* the service; it stores the website and responds to client requests (e.g., the e-commerce web server).
Request & Response	The client sends a **request** (e.g., "GET the home page") and the server returns a **response** containing a status code (200 OK, 404 Not Found) and the content (HTML, images, data).
HTTP	HyperText Transfer Protocol: the set of rules for exchanging web pages. Data is sent as **plain text**, so it can be read or modified by anyone intercepting it. Uses port 80.
HTTPS	HTTP over TLS/SSL: HTTP with **encryption**, authentication of the server using a digital certificate, and integrity checking. Uses port 443. The padlock in the browser shows HTTPS.
DNS Lookup	The process of converting the domain name to the server's IP address (see Task 3.6).
Packet Transmission	Data is broken into small units called packets. Each packet carries source and destination IP addresses and a sequence number, travels through routers independently, and is reassembled at the destination (TCP/IP).

4.2 Step-by-Step Process

Step 1: User enters URL. The user types `https://www.amazon.in` in the browser.

Step 2: DNS lookup. The browser checks its cache; if not found, the DNS resolver finds the IP address of the domain.

Step 3: Connection set-up. The browser opens a TCP connection to the server using a three-way handshake (SYN, SYN-ACK, ACK). For HTTPS, a TLS handshake follows, in which the server presents its certificate and both sides agree on encryption keys.

Step 4: HTTP request. The browser sends an HTTP request (e.g., `GET / HTTP/1.1`). The request is divided into packets and sent through the local router and ISP across many routers.

Step 5: Server processing. The web server (with application and database servers behind it) processes the request, e.g., fetches products and user data.

Step 6: HTTP response. The server sends back a response with a status code and the page content, again as packets.

Step 7: Rendering. The browser reassembles the packets, decrypts them, and displays the web page. Further requests are made for images, scripts, and styles.

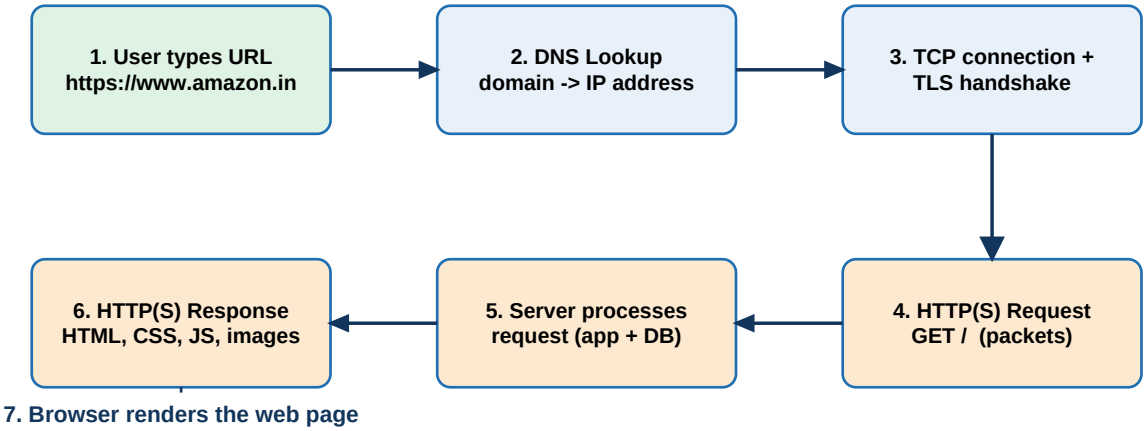

Figure 5: Flow diagram of communication between a client and a website

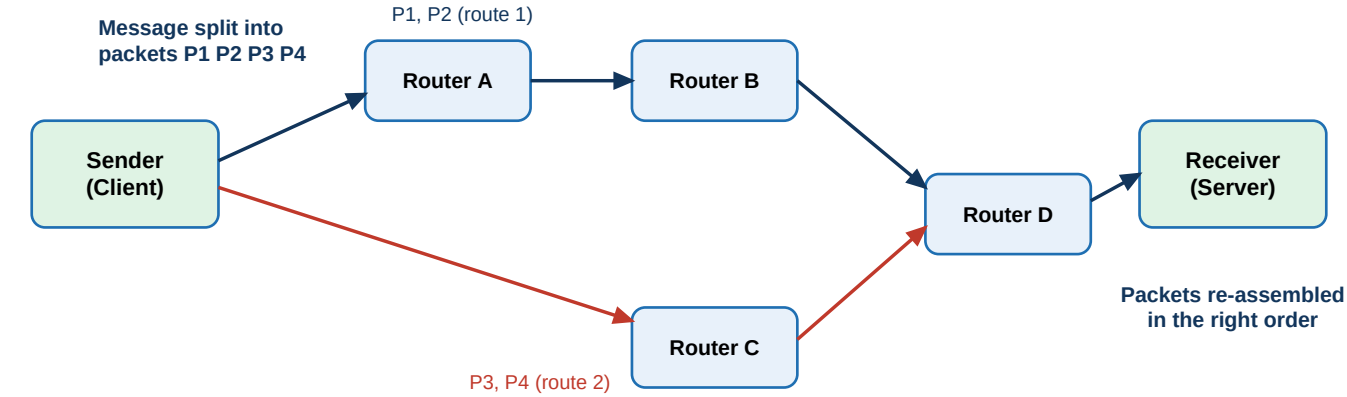

Figure 6: Packet transmission: a message is split, routed independently, and reassembled

4.3 HTTP vs HTTPS

Feature	HTTP	HTTPS
Security	No encryption; data visible in plain text	Encrypted with TLS; data is protected
Port	80	443
Certificate	Not required	Required (issued by a Certificate Authority)
Suitable for	Non-sensitive content (rarely used today)	Logins, payments, and all modern websites
Browser display	"Not secure" warning	Padlock icon

Task 5: Network Security Basics

Network security uses multiple layers of protection (defence in depth), so that if one control fails, others still protect the system. The five core controls are explained below.

5.1 Firewall

A **firewall** is a hardware or software security system that monitors and filters incoming and outgoing network traffic based on a set of security rules. It acts as a barrier between a trusted internal network and an untrusted network such as the Internet.

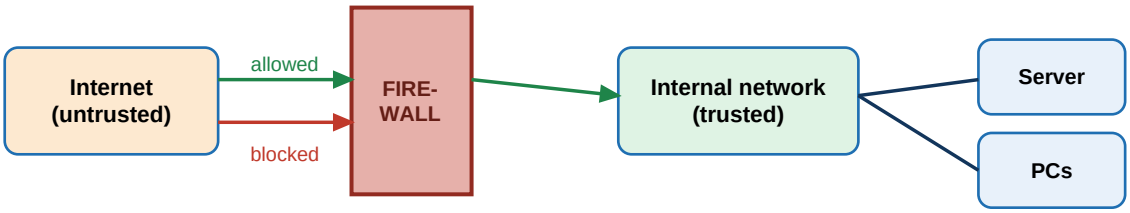

Figure 7: A firewall between the Internet and an internal network

How it works	Inspects packets (source/destination IP, port, protocol) and allows or blocks them according to rules. Types include packet-filtering, stateful inspection, application-layer (proxy), and Next-Generation Firewalls (NGFW).
Why important	Stops unauthorised access and many attacks, controls which services are reachable, and blocks suspicious traffic before it reaches servers.
E-commerce use	Allows only web traffic (ports 80/443) to the web servers and blocks direct Internet access to the database.

5.2 Antivirus

Antivirus software detects, blocks, and removes malicious software such as viruses, worms, trojans, spyware, and ransomware from devices.

How it works	Scans files and programs using signature matching (known malware patterns), heuristic/behaviour analysis (suspicious actions), and cloud intelligence; quarantines or deletes threats. Needs frequent updates.
Why important	Provides the first line of defence on endpoints, prevents infection and data loss, and detects threats that users may not notice.
E-commerce use	Installed on employees' computers and servers to prevent malware from reaching internal systems.

5.3 VPN (Virtual Private Network)

A **VPN** creates a secure, encrypted tunnel between your device and a VPN server over the Internet, hiding your traffic and IP address from outsiders.

How it works	The VPN client encrypts all data before it leaves the device and sends it through the tunnel to the VPN server, which decrypts it and forwards it to the destination. Common protocols include OpenVPN, IPsec, and WireGuard.

Why important	Protects data on public Wi-Fi, enables secure remote access to company networks, and improves privacy by masking the user's real IP address.
E-commerce use	Employees working remotely use a corporate VPN to reach internal admin tools safely.

5.4 Encryption

Encryption converts readable data (plaintext) into unreadable code (ciphertext) using an algorithm and a key. Only someone with the correct key can decrypt it.

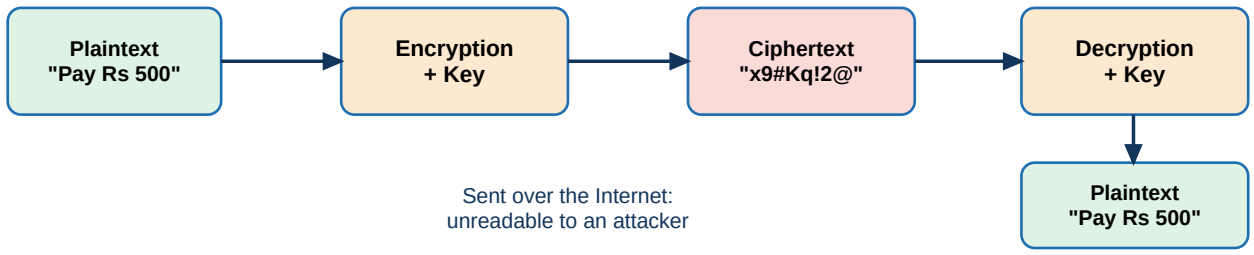

Figure 8: Encryption and decryption process

Types	**Symmetric** (same key for both, e.g., AES: fast) and **Asymmetric** (public/private key pair, e.g., RSA, ECC: used for key exchange and digital signatures). HTTPS uses both.
Why important	Protects confidentiality of data in transit (HTTPS/TLS) and at rest (encrypted databases and disks). Even if data is stolen, it remains unreadable.
E-commerce use	Card details and passwords are encrypted during checkout and stored securely (passwords are hashed, not stored as plain text).

5.5 Authentication

Authentication is the process of verifying that a user or device is who they claim to be before granting access. It is different from *authorisation*, which decides what the verified user is allowed to do.

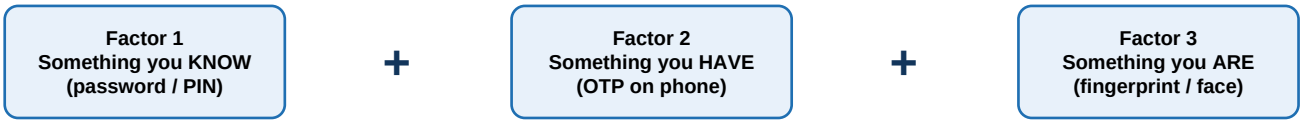

Figure 9: Authentication factors; using two or more is called Multi-Factor Authentication (MFA)

Methods	Passwords/PINs, OTPs, authenticator apps, biometrics (fingerprint, face), smart cards, and digital certificates.
Why important	Prevents unauthorised access and account takeover. MFA keeps accounts safe even if a password is stolen.
E-commerce use	Login with password + OTP for customers; strong MFA and role-based access for administrators and sellers.

5.6 Summary Table

Control	Main purpose	CIA goal supported
Firewall	Filters network traffic	Confidentiality, Availability
Antivirus	Detects and removes malware	Integrity, Availability
VPN	Secure remote communication	Confidentiality
Encryption	Makes data unreadable to attackers	Confidentiality, Integrity
Authentication	Verifies identity	Confidentiality, Integrity

Task 6: Practical Activity

Option selected: Option A. A basic office network diagram for a small e-commerce seller / startup office, with all devices labelled.

6.1 Network Diagram

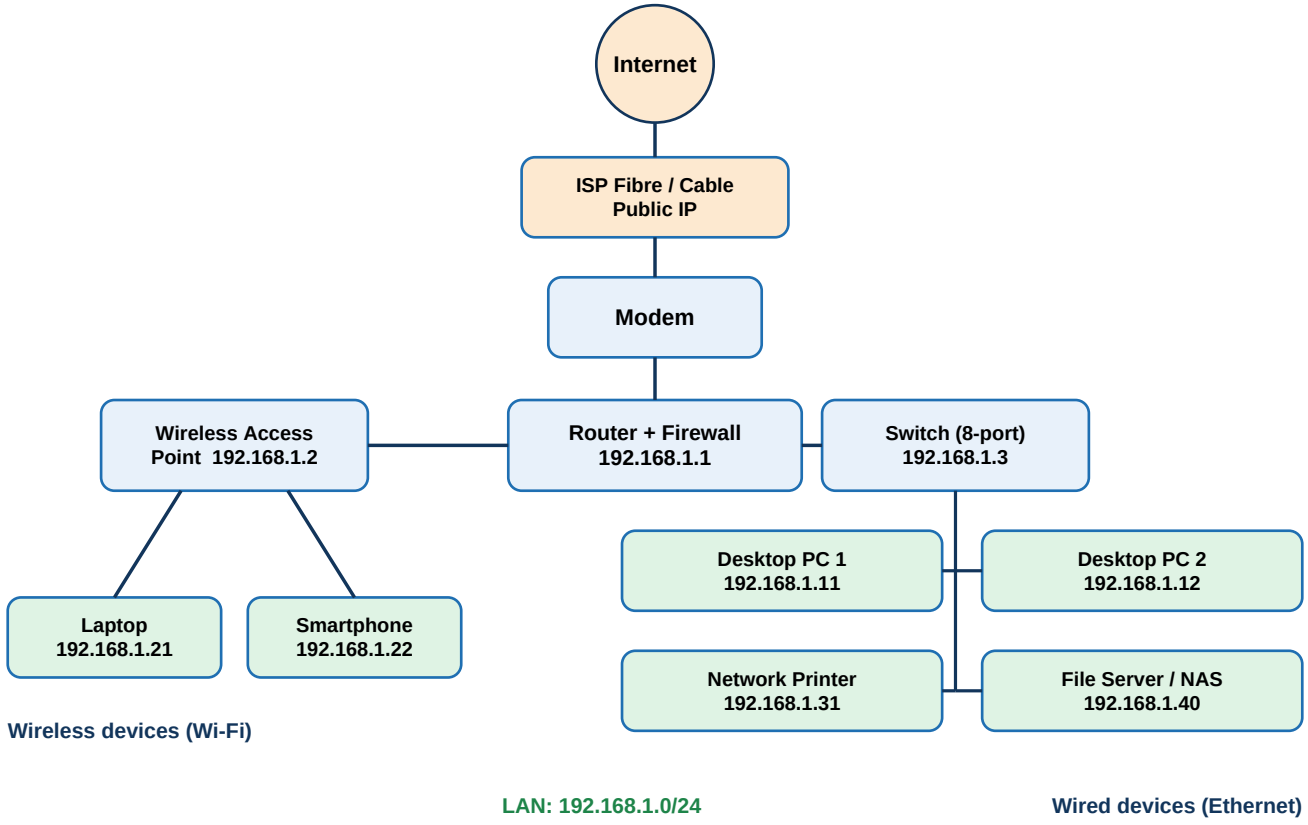

Figure 10: Basic office network diagram with labelled devices and private IP addresses

6.2 Device Description

Device	IP Address	Role in the network
Internet / ISP	Public IP (assigned by ISP)	Provides Internet connectivity to the office.
Modem	N/A (bridge mode)	Converts the ISP signal into digital data for the router.
Router + Firewall	192.168.1.1 (gateway)	Connects the LAN to the Internet, performs NAT and DHCP, and filters traffic with firewall rules.
Wireless Access Point	192.168.1.2	Provides Wi-Fi to laptops and smartphones.
Switch (8-port)	192.168.1.3	Connects wired devices in the LAN using MAC addresses.
Desktop PC 1 & 2	192.168.1.11 / .12	Staff computers for order processing and customer support.
Laptop & Smartphone	192.168.1.21 / .22	Wireless devices of employees.
Network Printer	192.168.1.31	Shared printer for invoices and shipping labels.
File Server / NAS	192.168.1.40	Central storage for documents and backups.

6.3 Explanation and Security Measures

Data from the Internet enters through the modem, which passes it to the router. The router uses the firewall to filter traffic and assigns private IP addresses (192.168.1.x) to all devices using DHCP. The switch connects the wired devices, while the access point serves wireless devices. All devices share one Internet connection but remain protected behind the router's NAT and firewall, since private addresses are not directly reachable from the Internet.

- Change default router/admin passwords and use WPA3 (or WPA2) encryption for Wi-Fi.
- Keep router firmware and device operating systems updated.
- Enable the firewall and disable unused services and ports.
- Use a separate guest Wi-Fi network for visitors.
- Install antivirus on all computers and take regular backups of the NAS.
- Use a VPN for remote access to the office network.

Note: Option B (system IP and screenshots) was not selected, so no screenshots are required for Option A. The IP addresses above are sample private addresses from the 192.168.0.0/16 range, used for illustration.

Conclusion

Cyber security is essential for every online organisation. This report explained the CIA Triad, the most common cyber threats, the basics of computer networks, how data travels from a client to a web server using DNS, HTTP/HTTPS, and packets, and the key defences: firewalls, antivirus, VPNs, encryption, and authentication. Using an e-commerce platform as the case study showed that no single control is enough. Strong security comes from **layered defence**, regular updates, user awareness, and continuous monitoring. A solid understanding of networking fundamentals is the foundation for learning advanced topics such as ethical hacking, network defence, and incident response.

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